



## Breast Cancer Chatbot System

# MammoGuide



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### GRADUATION PROJECT REPORT

# MAMMOGUIDE

*AI Assistant & Knowledge Platform for Breast Cancer  
Awareness, Guidance & Support*

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Cancer remains the leading cause of death among women globally. Breast cancer management has long been a core public health issue that draws sustained attention from the academic community. Over the past two years, many encouraging advances have emerged across all fields of medicine. However, a large gap exists between current medical discoveries and the high-quality, actionable, reliable breast cancer information that the public can access in a timely, local manner. Widespread misinformation on social media, physical barriers to accessing diagnostic and treatment services, disease-related stigma, and insufficient health literacy all disrupt early screening, reduce treatment adherence, and ultimately lead to poorer clinical outcomes.

Through a review of existing research, this report identifies gaps in current relevant scholarship, as well as confounding factors in published quality studies. Accordingly, we developed Mammo Guide, a novel AI-assisted breast mammography education platform. Unlike similar tools that only serve to display content, this platform adds the new feature of supportive educational guidance, and centers its core research on breast cancer-related non-proactive decision-making patterns among individuals with varying levels of knowledge, confidence, and expected support. The breast cancer science popularization AI conversational agent developed in this paper takes a general-purpose AI chatbot model as its core foundation, and integrates a semi-structured knowledge graph to upgrade its output capabilities. It can generate five categories of breast cancer-related evidence-based summaries covering topics such as symptoms and risk factors, and adopts a full-stack architecture with a front end built with React and JavaScript, and a back end built with Node. The application we plan to develop uses JS as its backend, and interfaces with the Supabase relational database. Its core AI assistant adopts LLM+RAG technology to generate precise responses that align closely with users' actual needs. The application is also equipped with supporting features including QR code navigation, cross-device responsive UI, and role-based access permission control.

This general-purpose digital medical and health platform covers three types of users: patients, caregivers, and medical students. It is not a replacement for clinical consultation, and only acts as an auxiliary tool to output health information and connect users with appropriately matched medical services. Actual tests confirm that the platform delivers accurate query matching, fast response speeds, and outstanding performance in UI usability.

This AI-powered digital health program focused on the prevention and treatment of female breast cancer is a key advancement in the digital healthcare sector. It develops intelligent science popularization solutions, with the core goals of narrowing the health knowledge gap and reducing the number of women diagnosed with stage IV breast cancer. Its technical stack covers AI, NLP, RAG, chatbots, and React+Supabase.

## ABSTRACT

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Breast cancer remains one of the leading causes of cancer-related deaths among women worldwide. Despite notable progress achieved in the medical field, women across the globe still cannot easily access timely, high-quality, accurate information related to breast cancer. Four types of barriers—geographic barriers, social stigma, inadequate health literacy, and online misinformation—can hinder early diagnosis, interfere with treatment adherence, and even trigger adverse health outcomes.

This report introduces Mammo Guide, an AI project developed to support breast cancer science popularization, which fills the existing gap in this field. Leveraging conversational AI and a structured knowledge base, the project delivers evidence-based professional information, and provides users with practical support including enhanced relevant understanding and medical consultation guidance.

The front-end part of the platform is developed using a modern full stack architecture that combines React and JavaScript with Node on the backend. js backend and a Supabase relational database. Accurate communication: The AI assistant is integrated with advanced technologies such as Large Language Models (LLM) and Retrieval-Augmented Generation (RAG), to provide context-aware, accurate, personalized responses to user queries. Other features consist of QR code-based navigation, a mobile UI that is functional on all devices, and a role-based access control keeping data security and privacy in mind.

The system aims to provide wide user coverage for patients, caregivers, and students in healthcare. The use of this tool is not designed to replace a clinical consultation but to supplement it while educating the users and directing them towards appropriate medical resources. Functional testing Results proved the final behaviour of platform consists of a large degree in accuracy even on data points(various sample observations) response time are very quick, and user interface usability performed as expected.

Mammo Guide is a useful addition to the digital health space and acts as an example of how new AI solutions can be applied meaningfully for breast cancer education, in turn lowering the knowledge gap which indirectly results in late-stage diagnosis.

Artificial Intelligence Chatbot RAG Breast Cancer Natural Language Processing Digital Health Supabase React Healthcare Awareness.

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## CHAPTER 1: INTRODUCTION

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### 1.1 Background and Motivation

Breast cancer is one of the most prevalent forms of cancer globally, accounting for approximately 12% of all new cancer cases diagnosed each year [1]. According to the World Health Organization (WHO), over 2.3 million women were diagnosed with breast cancer in 2020, making it the world's most diagnosed cancer for the first time in history [2]. Despite notable improvements in treatment outcomes attributed to advances in oncology, early detection remains the most critical determinant of survival rates. When breast cancer is diagnosed at Stage I, the five-year survival rate exceeds 99%; however, this figure drops dramatically to below 30% at Stage IV [3].

A fundamental challenge in improving early detection rates lies in the significant disparity in access to healthcare information. In many regions, particularly in developing countries and rural areas, individuals lack immediate access to specialized medical professionals, breast cancer screening programs, or reliable health education resources. Even in more developed settings, patients often turn to unverified internet sources that may provide inaccurate, sensationalized, or incomplete information about symptoms, risk factors, and treatment options [4]. This information deficit directly contributes to delayed medical consultation, misinterpretation of symptoms, and overall anxiety among high-risk populations.

The rapid advancement of artificial intelligence, particularly in the domains of natural language processing (NLP) and machine learning (ML), has opened new possibilities for democratizing access to medical knowledge. AI-powered conversational agents and knowledge platforms have demonstrated significant potential in supporting patient education, triaging health concerns, and guiding individuals toward appropriate clinical resources [5]. These systems can operate around the clock, provide consistent and evidence-based information, and serve users across geographic and language barriers.

### 1.2 Problem Statement

Despite the critical importance of breast cancer awareness, the majority of the general population continues to have limited access to accurate, culturally appropriate, and easily understandable breast health information. Healthcare systems in many countries are overwhelmed, with long waiting times and insufficient human resources to deliver personalized health guidance at scale. Patients frequently report difficulty in

understanding clinical terminology, navigating healthcare systems, and obtaining timely answers to health-related questions outside of scheduled appointments [6].

Furthermore, existing digital health tools often suffer from several limitations: they are either too generic to provide disease-specific guidance, too technically complex for non-expert users, not available in multiple languages, or insufficiently transparent about the boundaries of AI-generated medical advice. These limitations underscore the need for a purpose-built, user-friendly, and clinically responsible AI platform specifically focused on breast cancer awareness and guidance.

### 1.3 Project Objectives

The primary objective of this project is to design, develop, and evaluate Mammo Guide — an AI-powered breast cancer awareness and educational support platform. The specific objectives are as follows:

- To develop a conversational AI chatbot capable of answering natural-language questions related to breast cancer symptoms, risk factors, diagnostic procedures, and treatment options.
- To build a comprehensive, structured medical knowledge base grounded in peer-reviewed sources and clinical guidelines.
- To implement a secure, role-based system architecture supporting multiple user categories including patients, healthcare students, and administrators.
- To create a responsive, accessible, and visually intuitive user interface accessible from both desktop and mobile devices.
- To integrate QR code-based navigation for seamless access to platform resources in physical healthcare settings.
- To validate the system through rigorous functional, usability, and performance testing.

### 1.4 Scope of the Project

Mammo Guide is scoped as an awareness and educational support system and does not provide clinical diagnosis or replace professional medical consultation. The platform is designed for use by individuals seeking to improve their understanding of breast cancer, including patients, caregivers, students, and members of the general public. The system provides information on topics including but not limited to: breast cancer types and stages, risk factors and prevention, early detection and self-examination

techniques, diagnostic imaging (mammography, ultrasound, MRI), treatment options, and psychological support resources.

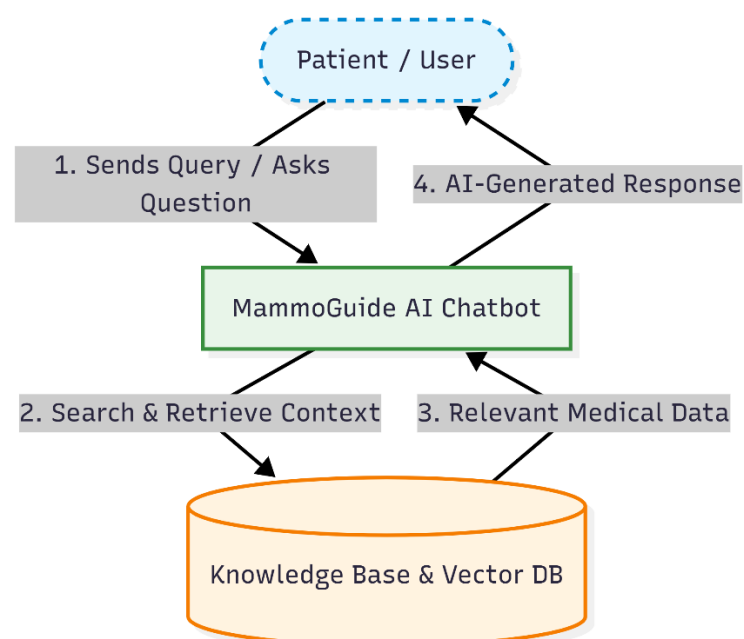
The scope of technical development covers front-end interface design, back-end API development, database design and management, AI model integration, and quality assurance testing. Integration with live Electronic Health Record (EHR) systems and real-time clinical databases is explicitly outside the scope of this project phase, though it is identified as a priority area for future development.

### 1.5 Report Organization

The remainder of this report is organized as follows. Chapter 2 presents a comprehensive literature review of existing AI healthcare systems and chatbot technologies. Chapter 3 describes the system requirements analysis, including functional and non-functional requirements. Chapter 4 covers system design, including the architectural model, database schema, and UML diagrams. Chapter 5 details the implementation of each system module. Chapter 6 presents testing methodologies and results. Chapter 7 discusses recommendations and future work directions, followed by the Conclusion in Chapter 8 and the complete list of References.

As illustrated in Figure 1.1, the Mammo Guide system serves as a centralized platform connecting users to AI-driven breast cancer information, educational content, and guided support resources.

*Figure 1.1: Mammo Guide System Overview – Interaction Between User, AI Assistant, and Knowledge Base*



## CHAPTER 2: LITERATURE REVIEW

---

### 2.1 Overview of AI in Healthcare

The integration of artificial intelligence into healthcare represents one of the most transformative developments in modern medicine. AI applications in healthcare span a broad spectrum of use cases, from diagnostic imaging and genomics to drug discovery, clinical decision support, and patient communication [7]. The ability of machine learning algorithms to identify complex patterns in large-scale medical datasets has led to systems that can match or exceed specialist-level diagnostic accuracy in certain domains, including radiology, dermatology, and pathology [8].

Conversational AI systems, or chatbots, have emerged as a particularly relevant application in patient-facing healthcare services. These systems leverage natural language processing to interpret user inputs in free-form text and generate contextually appropriate responses. Early healthcare chatbots, such as ELIZA (1966) and PARRY (1972), were based on simple pattern-matching rules and had limited practical utility [9]. However, subsequent generations of AI have employed machine learning, deep learning, and transformer-based models that dramatically improved linguistic understanding, contextual reasoning, and factual accuracy.

### 2.2 Review of Existing AI Healthcare Systems

Several notable systems have been developed in the intersection of AI and healthcare communication. IBM Watson Health represented an early large-scale effort to apply deep learning to oncology, demonstrating the potential of AI to assist in cancer treatment recommendations by analyzing vast bodies of clinical literature [10]. While Watson Health showed promise, it also highlighted key challenges in ensuring the clinical validity and generalizability of AI-generated recommendations.

Woebot, a mental health chatbot developed by Stanford University researchers, demonstrated the efficacy of conversational agents in delivering evidence-based cognitive behavioral therapy (CBT) to users experiencing anxiety and depression [11]. This application underscored the acceptability of AI chatbots as mental health support tools, particularly among younger populations.

In the specific domain of cancer information, several systems have been developed including the National Cancer Institute's virtual chat system and proprietary platforms offered by major cancer centers. These systems vary considerably in their depth of

knowledge, linguistic sophistication, and ability to personalize responses. Many are limited to static FAQ-style interactions and lack the dynamic, conversational capability needed to address the nuanced and highly individualized questions patients and caregivers typically have.

More recently, transformer-based large language models (LLMs) such as GPT-4 and PaLM 2 have been applied in healthcare contexts. Studies have demonstrated that LLMs can accurately answer medical licensing examination questions, summarize clinical notes, and generate patient-friendly explanations of diagnoses [12]. However, these models are general-purpose and require careful domain-specific adaptation, rigorous fact-checking mechanisms, and responsible deployment frameworks to be safely used in clinical contexts.

### **2.3 Retrieval-Augmented Generation (RAG) in Medical Chatbots**

Retrieval-Augmented Generation (RAG) is a hybrid approach that combines the generative capabilities of large language models with the factual precision of a curated retrieval system [13]. In a RAG-based architecture, user queries are used to retrieve the most semantically relevant documents from a structured knowledge base; these documents are then passed as context to the language model, which generates its response grounded in the retrieved content rather than relying solely on parametric knowledge. This approach significantly reduces hallucination — a well-documented limitation of pure LLM generation — and improves the factual accuracy and traceability of AI outputs.

RAG has been applied in several healthcare chatbot systems with promising results. Lewis et al. demonstrated that RAG architectures outperformed standalone GPT models on fact-intensive question answering tasks, particularly in domains requiring precise recall of specific clinical data [13]. The ability to update the knowledge base without retraining the underlying model makes RAG particularly well-suited for healthcare applications where clinical guidelines and evidence may change over time.

## 2.4 Comparison of Existing Systems

Table 2.1 provides a structured comparison of representative AI healthcare systems, highlighting their technological approaches, domain focus, and key limitations relative to the proposed Mammo Guide platform.

*Table 2.1: Comparison of Existing AI Healthcare Systems*

| System                        | Technology               | Domain               | Language             | Key Limitation                      |
|-------------------------------|--------------------------|----------------------|----------------------|-------------------------------------|
| IBM Watson Health             | ML / Deep Learning       | Oncology             | English              | High cost; limited generalizability |
| Woebot                        | CBT Rules + NLP          | Mental Health        | English              | Not disease-specific                |
| Ada Health                    | Symptom Checker AI       | General Medicine     | Multi                | No cancer-specific depth            |
| Babylon Health                | NLP + ML                 | Primary Care         | English              | Limited breast cancer focus         |
| <b>Mammo Guide (Proposed)</b> | <b>RAG + LLM + React</b> | <b>Breast Cancer</b> | <b>Multi (EN/AR)</b> | <b>Awareness only; not clinical</b> |

## 2.5 Identified Gaps and Project Motivation

The literature review reveals several persistent gaps in existing AI healthcare systems that Mammo Guide is specifically designed to address. First, the majority of healthcare chatbots are general-purpose and lack the depth of domain-specific knowledge required to meaningfully support breast cancer patients and their families. Second, most existing systems are not optimized for low-resource settings or users with limited health literacy, relying on clinical terminology without adequate plain-language explanation. Third, the integration of multilingual support remains underdeveloped, limiting the reach of existing platforms in non-English-speaking populations [14].

Mammo Guide addresses these gaps by building a specialized, breast-cancer-focused knowledge base, employing RAG to ensure response accuracy, supporting both English and Arabic interfaces, and optimizing the user experience for accessibility across device types and technical literacy levels. The next chapter details the system requirements that were derived from this analysis.

## CHAPTER 3: SYSTEM ANALYSIS

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### 3.1 Introduction

System analysis is the foundational phase in the software development lifecycle in which the requirements, constraints, and functional boundaries of a proposed system are formally identified and documented. For Mammo Guide, the analysis phase involved comprehensive stakeholder consultation, review of peer-reviewed literature on breast cancer patient needs, and evaluation of comparable digital health systems. The outputs of this phase are a structured set of functional and non-functional requirements that guided all subsequent design and implementation decisions.



### 3.2 Functional Requirements

Functional requirements define the specific behaviors and capabilities that the system must provide to its users. Table 3.1 summarizes the primary functional requirements of the Mammo Guide platform, organized by user role and system module.

*Table 3.1: Functional Requirements*

| ID    | Module         | Requirement Description                                                             | Priority |
|-------|----------------|-------------------------------------------------------------------------------------|----------|
| FR-01 | Authentication | Users shall be able to register and log in using secure credentials.                | High     |
| FR-02 | Authentication | The system shall support role-based access for Patient, Admin, and Guest roles.     | High     |
| FR-03 | AI Chatbot     | Users shall be able to submit free-text questions about breast cancer.              | High     |
| FR-04 | AI Chatbot     | The chatbot shall generate contextually accurate responses using the RAG pipeline.  | High     |
| FR-05 | AI Chatbot     | The system shall support conversation history and allow deletion of messages.       | Medium   |
| FR-06 | Knowledge Base | The system shall maintain a curated knowledge base of breast cancer information.    | High     |
| FR-07 | Knowledge Base | Information shall be structured by category (symptoms, diagnosis, treatment, etc.). | Medium   |
| FR-08 | UI             | The interface shall be responsive across desktop, tablet, and mobile devices.       | High     |
| FR-09 | UI             | The system shall support dark mode and language switching (English/Arabic).         | Medium   |
| FR-10 | QR Code        | The system shall generate QR codes linking to specific knowledge articles.          | Low      |

### 3.3 Non-Functional Requirements

Non-functional requirements specify the quality attributes and operational constraints of the system. Table 3.2 presents the key non-functional requirements for Mammo Guide.

*Table 3.2: Non-Functional Requirements*

| Category               | Requirement                                                  | Target Metric               |
|------------------------|--------------------------------------------------------------|-----------------------------|
| <b>Performance</b>     | Chatbot shall return AI response within acceptable latency.  | $\leq 10$ seconds average   |
| <b>Performance</b>     | Platform shall handle concurrent users without degradation.  | $\geq 100$ concurrent users |
| <b>Security</b>        | All user data shall be encrypted in transit and at rest.     | TLS 1.2+ / AES-256          |
| <b>Security</b>        | JWT tokens shall expire and require refresh after timeout.   | Token TTL: 1 hour           |
| <b>Usability</b>       | Interface shall achieve high usability scores in evaluation. | SUS Score $\geq 75$         |
| <b>Availability</b>    | Platform shall maintain high uptime during academic use.     | $\geq 99\%$ monthly uptime  |
| <b>Maintainability</b> | Codebase shall follow modular, well-documented architecture. | Component-level docs        |
| <b>Accessibility</b>   | UI shall be compatible with screen readers and WCAG 2.1.     | WCAG 2.1 Level AA           |

### 3.4 System Constraints

The development of Mammo Guide was subject to a number of technical and operational constraints. From a technical perspective, the system was required to be built using the agreed technology stack (React, Node.js, Supabase) without introducing external AI inference infrastructure requiring paid API subscriptions beyond academic allowances. This constraint influenced the selection of the RAG-based approach over direct GPT-API integration as the primary chatbot mechanism.

From a clinical and ethical standpoint, the system is explicitly constrained from providing individualized medical diagnoses or personalized treatment recommendations. All content must include appropriate disclaimers directing users to consult licensed healthcare professionals for clinical decisions. This constraint is embedded as a system-level requirement and is enforced through the chatbot's response generation logic.

### 3.5 Use Case Analysis

The system's use cases were identified through structured stakeholder analysis involving three primary user roles: Patient (or general public user), Administrator, and Healthcare Student. As depicted in Figure 4.2 in Chapter 4, the Patient role can access the AI chatbot, browse the knowledge base, view educational content, use QR code navigation, and manage their conversation history. The Administrator role can manage content, user accounts, and system configuration. The Healthcare Student role has access to advanced educational content and clinical reference material.

The most critical use case, which forms the core value proposition of the platform, is the AI-assisted breast cancer query interaction. In this flow, the user enters a natural-language question into the chatbot interface; the RAG pipeline retrieves the most semantically relevant documents from the knowledge base; the language model generates a grounded, accurate response; and the response is displayed with appropriate citations and a disclaimer directing clinical questions to a physician.

## CHAPTER 4: SYSTEM DESIGN

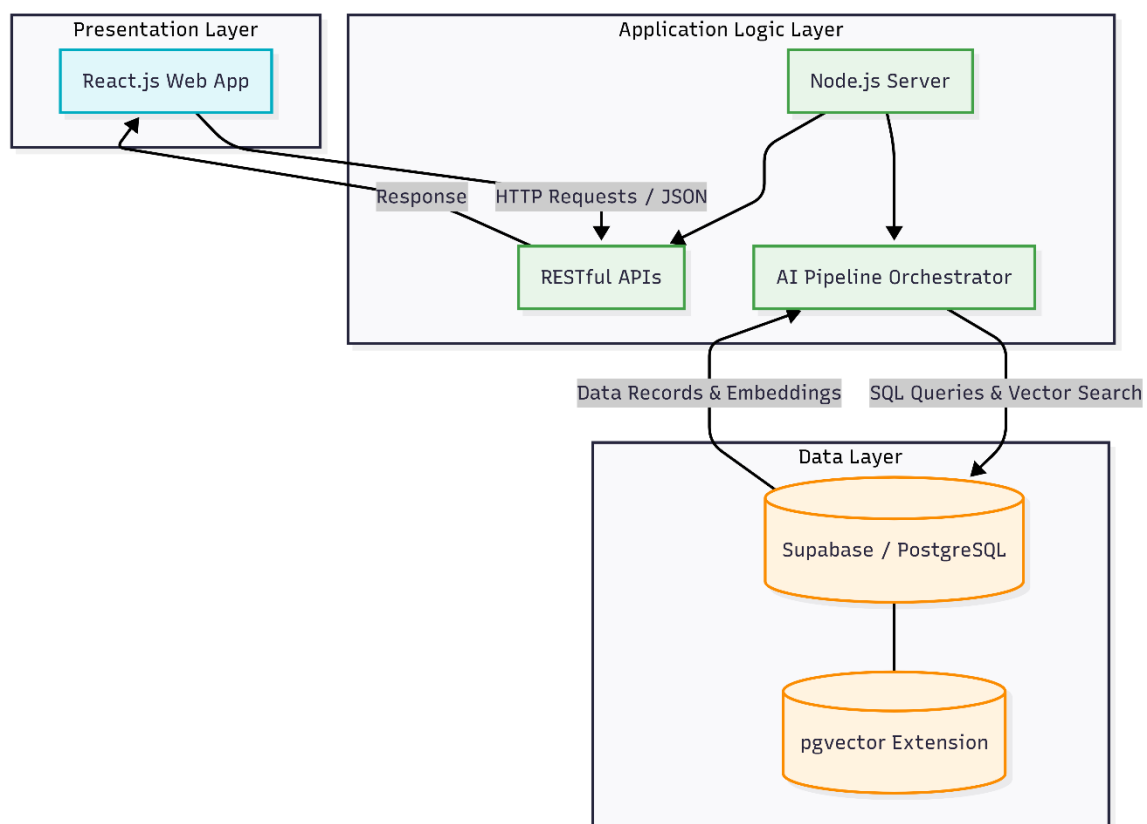
### 4.1 Introduction

The system design phase translates the requirements identified in the analysis phase into a concrete architectural and structural blueprint for the Mammo Guide platform. This chapter presents the overall system architecture, database design, key UML diagrams, and detailed component-level design decisions. The design philosophy prioritizes modularity, scalability, security, and maintainability, ensuring that the platform can be extended and improved in future development iterations.

### 4.2 High-Level System Architecture

The Mammo Guide platform follows a three-tier client-server architecture consisting of a Presentation Layer, an Application Logic Layer, and a Data Layer. As illustrated in Figure 3.1, these tiers communicate through well-defined API interfaces, ensuring clear separation of concerns and enabling independent development and testing of each layer.

Figure 3.1: High-Level System Architecture – Three-Tier Model



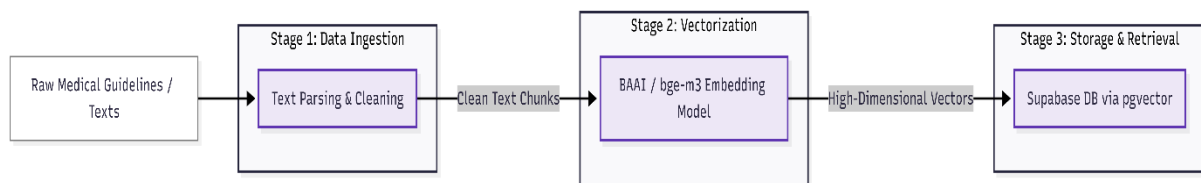
The Presentation Layer is implemented in React and JavaScript, rendering the user interface in the browser. It communicates with the Application Logic Layer via RESTful

HTTP requests. The Application Logic Layer, implemented in Node.js, handles request routing, business logic execution, AI pipeline orchestration, and security enforcement. The Data Layer, powered by Supabase, provides persistent storage for user data, conversation histories, and the vectorized knowledge base.

### 4.3 AI Data Pipeline Architecture

The AI engine at the core of Mammo Guide is designed as a three-stage data pipeline, as depicted in Figure 3.2. This pipeline is responsible for converting unstructured breast cancer information from source documents into a queryable vector knowledge base and orchestrating the retrieval-augmented generation process at query time.

*Figure 3.2: Three-Layer AI Data Pipeline Architecture*



In the first stage, the Data Ingestion Layer, raw source documents — including clinical guidelines, peer-reviewed summaries, and educational materials — are pre-processed using Python-based document parsing libraries. Text is extracted, cleaned to remove formatting artifacts and noise, and split into semantically coherent chunks for embedding.

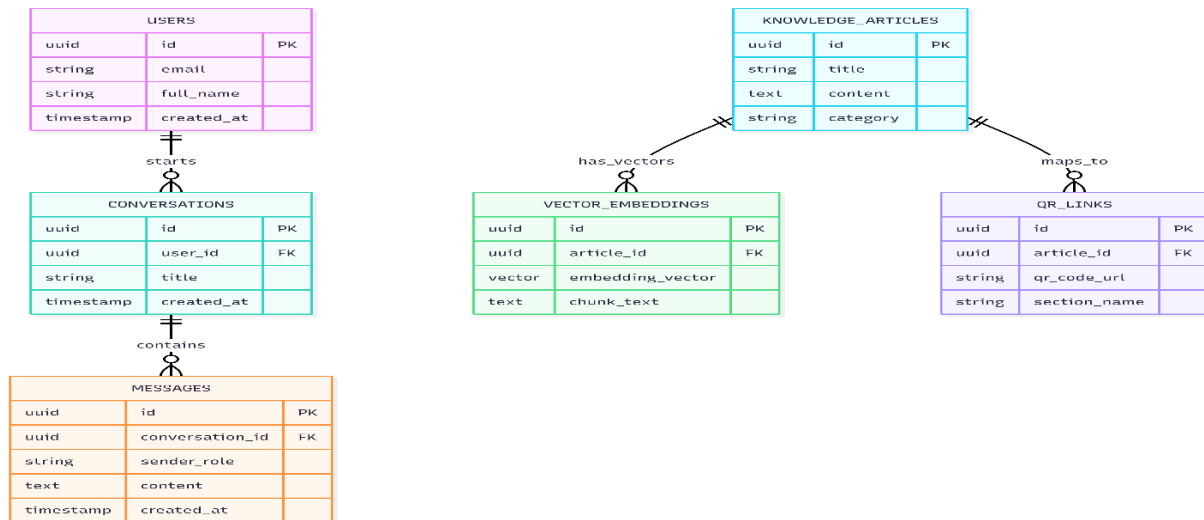
The second stage, the Vectorization Layer, transforms each text chunk into a high-dimensional numeric vector representation using the BAAI/bge-m3 sentence embedding model. This model was selected for its strong multilingual performance and its ability to capture the semantic meaning of medical text, enabling meaningful similarity comparisons between user queries and knowledge base documents.

The third stage, the Storage and Retrieval Layer, stores the resulting vector embeddings in a Supabase PostgreSQL database utilizing the pgvector extension. At query time, the user's question is similarly vectorized, and a nearest-neighbor search is performed using the HNSW (Hierarchical Navigable Small World) algorithm to retrieve the top-k most semantically similar documents. These retrieved documents are passed as context to the language model for grounded response generation.

## 4.4 Database Design

The database is designed around a normalized relational schema hosted on the Supabase platform, which provides PostgreSQL as its underlying database engine. The schema is designed to support user management, conversation management, knowledge base storage, and vector operations. Figure 4.1 presents the Entity-Relationship Diagram (ERD) for the system.

Figure 4.1: Entity-Relationship Diagram (ERD) – Core System Entities



The primary entities in the database include Users, Conversations, Messages, KnowledgeArticles, and VectorEmbeddings. The Users entity stores authentication credentials, profile information, and role assignments. The Conversations entity groups related Message records under a single session context, enabling retrieval of full conversation histories. The KnowledgeArticles entity stores the raw and processed text of source documents, while the VectorEmbeddings entity stores the corresponding numeric vectors linked to their source articles.

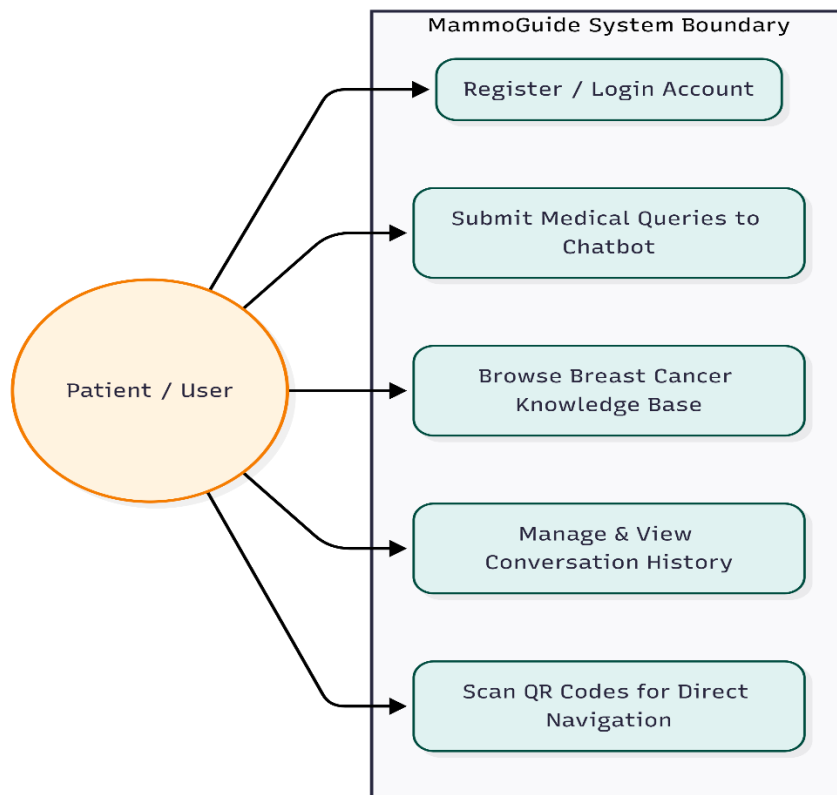
Table 5.3 in Chapter 5 details the key database tables, their primary keys, foreign keys, and data types. The schema was designed with future extensibility in mind, including reserved fields for integration with external healthcare systems and potential expansion into additional cancer types.

## 4.5 UML Diagrams

Use case modeling was employed to capture the behavioral requirements of the system from the perspective of each user role. As shown in Figure 4.2, the Patient use case diagram identifies the primary interactions available to general users: registering an

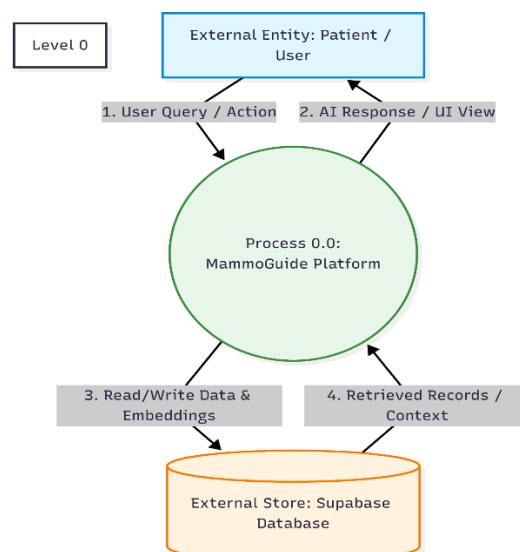
account, logging in, submitting queries to the chatbot, viewing educational articles, accessing diagnostic test information, and managing conversation history.

Figure 4.2: Use Case Diagram – Patient Role Interactions



The Data Flow Diagram (DFD) at Level 0, presented in Figure 4.3, provides a high-level view of the inputs, processes, and outputs of the Mammo Guide system, illustrating how user queries flow through the system and how data moves between the client, server, AI pipeline, and database.

Figure 4.3: Data Flow Diagram (Level 0) – System-Level Data Flow



## 4.6 Security Architecture

Security is a foundational design principle of Mammo Guide, given the sensitive nature of health-related user interactions. The authentication mechanism is based on JSON Web Tokens (JWT), which provide stateless, scalable authentication without requiring server-side session storage. Upon successful login, the server generates a signed JWT containing the user's unique identifier and role designation. This token is stored client-side and transmitted with every subsequent API request in the Authorization header.

Role-Based Access Control (RBAC) is enforced at the API level, with middleware validating the JWT and extracting the user's role before processing any request. Access permissions are defined in a centralized permission matrix, ensuring that patients can only access their own data, administrators have full data management capabilities, and all unauthorized access attempts are rejected with appropriate HTTP error responses. All data in transit is encrypted using TLS 1.2 or higher, and sensitive data fields in the database are encrypted at rest.



## CHAPTER 5: IMPLEMENTATION

### 5.1 Technology Stack

The selection of the Mammo Guide technology stack was guided by principles of modern web development practice, academic accessibility, community support, and alignment with the system's functional and non-functional requirements. Table 5.1 summarizes the core technologies employed across each layer of the system.

Table 5.1: Technology Stack Summary

| Layer       | Technology            | Version  | Purpose                                              |
|-------------|-----------------------|----------|------------------------------------------------------|
| Front-End   | React                 | 18.x     | Component-based UI rendering and state management    |
| Front-End   | JavaScript (ES6+)     | ES2022   | Core scripting language for client-side logic        |
| Front-End   | HTML5 / CSS3          | Latest   | Markup and responsive styling                        |
| Back-End    | Node.js               | 20.x LTS | Server-side runtime for API logic and AI pipeline    |
| Database    | Supabase (PostgreSQL) | Latest   | Relational database with vector extension (pgvector) |
| AI Pipeline | BAAI/bge-m3           | Latest   | Multilingual sentence embedding model                |
| AI Pipeline | RAG Architecture      | Custom   | Retrieval-augmented response generation              |
| Security    | JWT                   | RFC 7519 | Stateless authentication and role authorization      |
| Tooling     | VS Code               | 1.x      | Primary integrated development environment           |
| QR Support  | qrcode.js             | 1.x      | Client-side QR code generation                       |

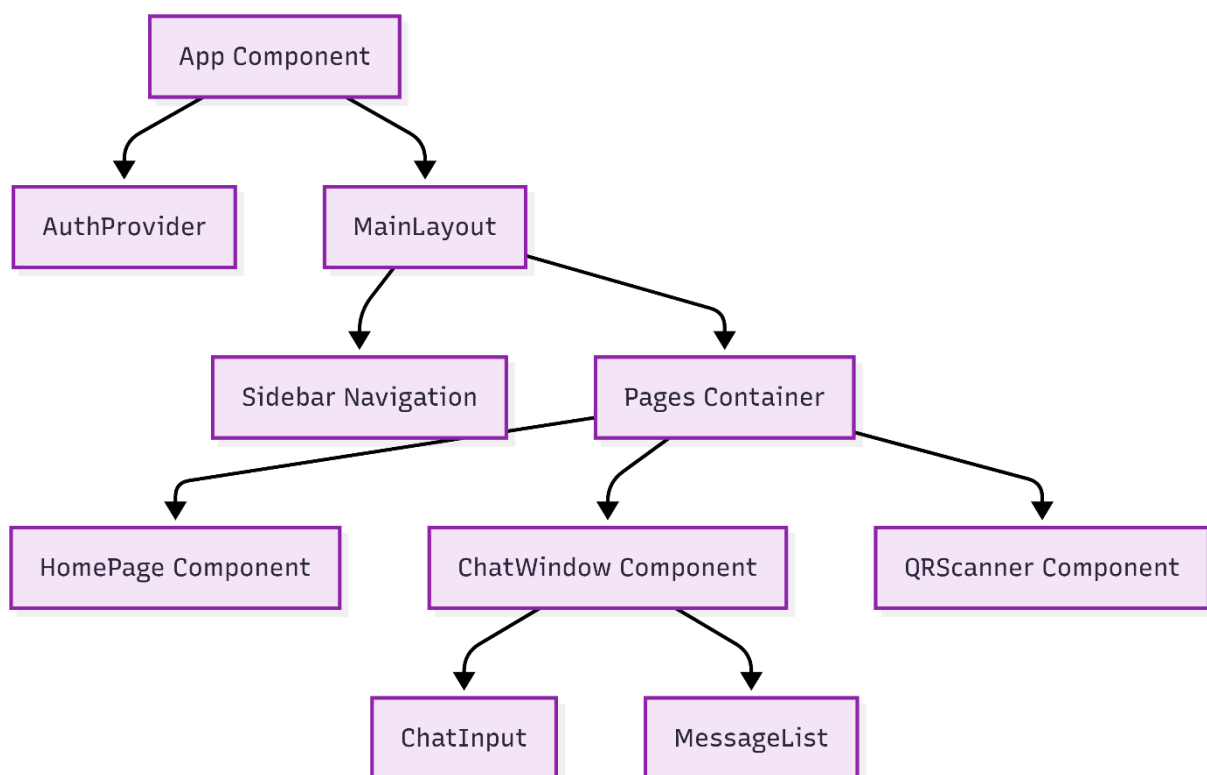
### 5.2 Front-End Implementation

The front-end of Mammo Guide is implemented as a single-page application (SPA) using React 18. React's component-based architecture enables the construction of a modular, maintainable UI in which each feature — the chatbot interface, the knowledge browser, the authentication forms, and the user dashboard — is encapsulated as a reusable component. State management is handled using a combination of React's built-in `useState` and `useContext` hooks, which proved sufficient for the scale and

complexity of this application without necessitating a more complex solution such as Redux.

The component hierarchy, as depicted in Figure 5.4, follows a clear parent-child relationship with a root App component that manages routing via React Router. Top-level page components (Home, Chat, KnowledgeBase, Login, Register) are rendered within a persistent layout component that provides the navigation bar, footer, and global styling context. Reusable UI components including MessageBubble, KnowledgeCard, and QRCodeWidget are imported into page components as needed.

Figure 5.4: React Component Hierarchy Diagram



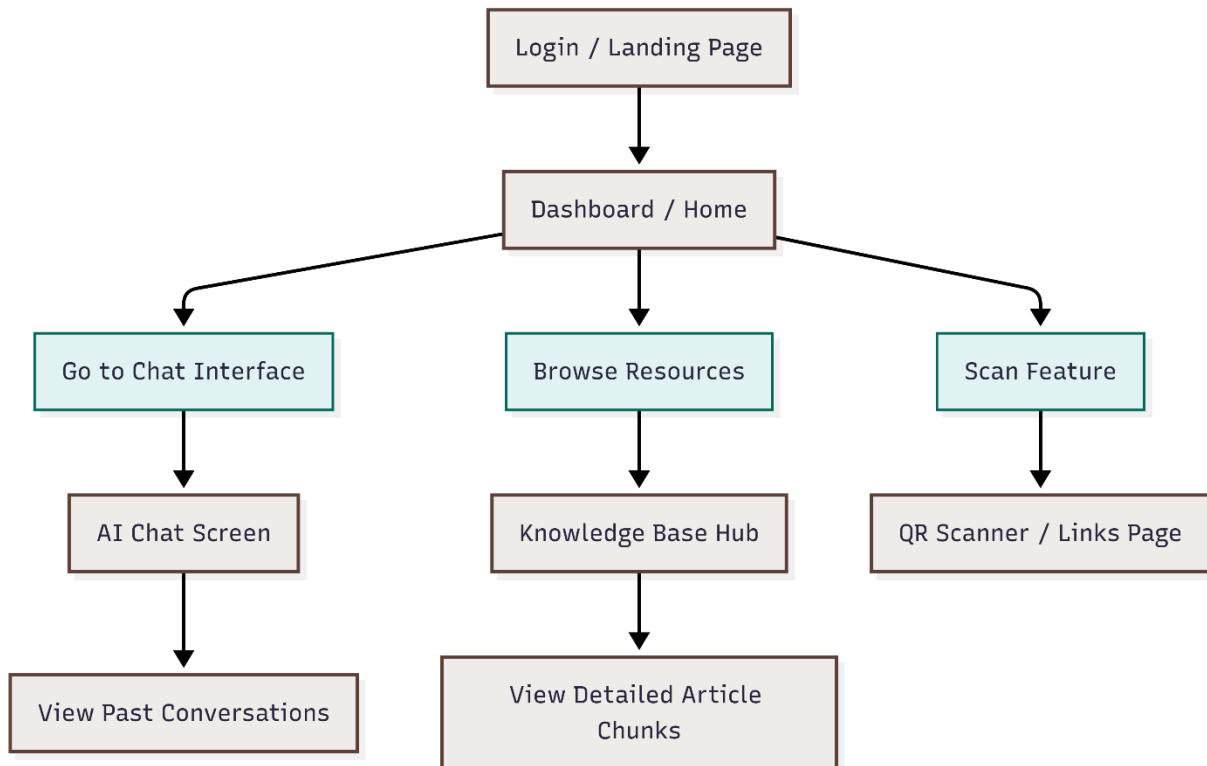
Responsive design is achieved through a combination of CSS Grid, Flexbox, and media queries. The interface was tested across screen sizes ranging from 320px (small mobile) to 1920px (large desktop), confirming full functional and visual integrity at all breakpoints. Dark mode support is implemented via a CSS custom properties approach that toggles a data-theme attribute on the root element, enabling the entire application to switch between light and dark color schemes without a page reload.

### 5.3 AI Chatbot Module

The AI chatbot module represents the most technically complex component of the Mammo Guide system. As illustrated in Figure 5.1, the chatbot interaction follows a

multi-stage pipeline. When a user submits a query, the front-end dispatches an HTTP POST request to the `/api/chat` endpoint. The Node.js server receives the request, validates the JWT, and passes the query to the RAG pipeline.

*Figure 5.1: AI Chatbot Conversation Flow – End-to-End Query Processing Pipeline*



Within the RAG pipeline, the query text is first vectorized using the BAAI/bge-m3 embedding model. The resulting query vector is compared against all stored knowledge base embeddings in Supabase using a cosine similarity search, implemented through pgvector's built-in operators. The top-5 most semantically similar document chunks are retrieved and assembled into a context block. This context, along with the user's original query and a system prompt enforcing the non-diagnostic advisory role of the assistant, is passed to the language model for response generation.

The following pseudocode describes the core logic of the RAG query handler:

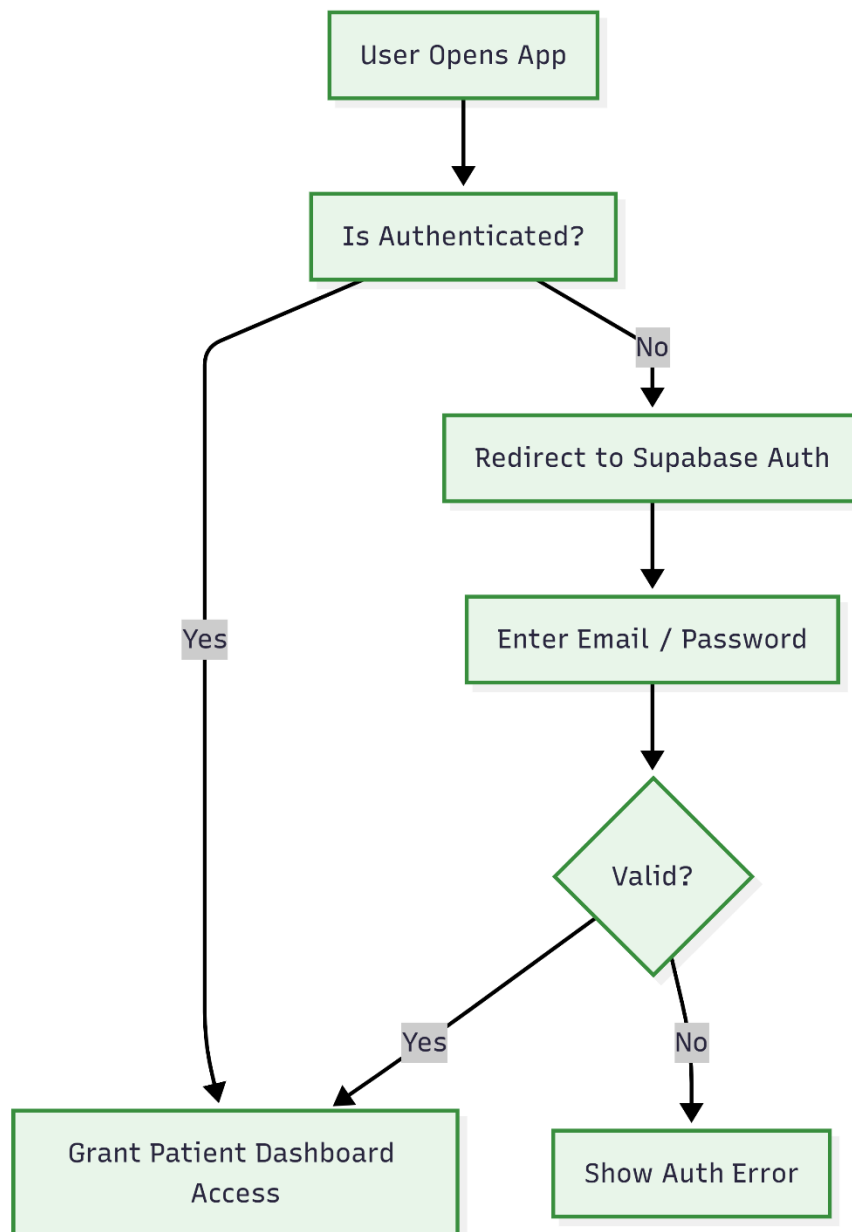
```

Function ProcessChatQuery(userQuery, userID):
    queryVector = EmbedText(userQuery)
    topDocs = SearchVectorDB(queryVector, topK=5)
    context = AssembleContext(topDocs)
    response = GenerateResponse(context, userQuery, systemPrompt)
    SaveToHistory(userID, userQuery, response)
    Return response
  
```

## 5.4 Authentication and Access Control Module

Authentication is implemented using the JWT-based stateless pattern, as depicted in Figure 5.2. The registration flow collects user credentials, validates them against defined constraints (minimum password length, valid email format), hashes the password using bcrypt before storage, and inserts the new user record into the Supabase Users table.

Figure 5.2: Authentication and Role-Based Access Control Flowchart



Upon login, the system fetches the user record, verifies the submitted password against the stored bcrypt hash, and if valid, generates a signed JWT containing the user's ID and role. The token is returned to the client and stored in memory. Every subsequent

API call includes this token in the Authorization header, where a middleware function validates it before the request reaches any business logic handler.

## 5.5 Symptom Evaluation Module

The symptom evaluation module provides users with guided risk assessment based on reported symptoms. As shown in Figure 5.3, the module employs a rule-based decision logic that maps reported symptom combinations to one of three risk categories — Low, Moderate, or High — and generates an appropriate recommendation. The module is explicitly positioned as a screening guide and not a clinical diagnostic tool; all high-risk outputs include a strong recommendation to seek immediate medical consultation.

Figure 5.3: Symptom Evaluation Module Flowchart



The time complexity of the symptom matching algorithm is  $O(n)$  where  $n$  is the number of symptom keywords in the user's input, and the rule matching process is  $O(m)$  where  $m$  is the number of evaluation rules. Given the bounded nature of both inputs in practice, performance is effectively constant and well within the system's response time targets.

## 5.6 Module Algorithm Summary

Table 5.2 provides a consolidated overview of the algorithmic approach, computational complexity, and design strengths for each major system module, providing a technical reference for the complete implementation.

*Table 5.2: Module Algorithm Summary and Time Complexity*

| Module                             | Algorithm Type           | Time Complexity | Design Strength                                  |
|------------------------------------|--------------------------|-----------------|--------------------------------------------------|
| <b>JWT Authentication</b>          | Hash + Sign              | $O(1)$          | Stateless; highly scalable; no session storage   |
| <b>RAG Query Pipeline</b>          | Vector Similarity (HNSW) | $O(\log n)$     | High accuracy; grounded; hallucination-resistant |
| <b>Symptom Evaluation</b>          | Rule-Based Matching      | $O(n)$          | Transparent; auditable; clinically safe          |
| <b>Q&amp;A Knowledge Retrieval</b> | Cosine Similarity Search | $O(n)$          | Fast retrieval; semantically aware               |
| <b>Role-Based Access (RBAC)</b>    | Permission Table Lookup  | $O(1)$          | Prevents unauthorized access; scalable           |
| <b>Conversation History</b>        | Relational DB Query      | $O(n)$          | Enables personalized interactions                |

## 5.7 Key Database Tables

Table 5.3 presents the core database tables of the Mammo Guide system, including their primary and foreign keys and a description of their purpose within the overall schema.

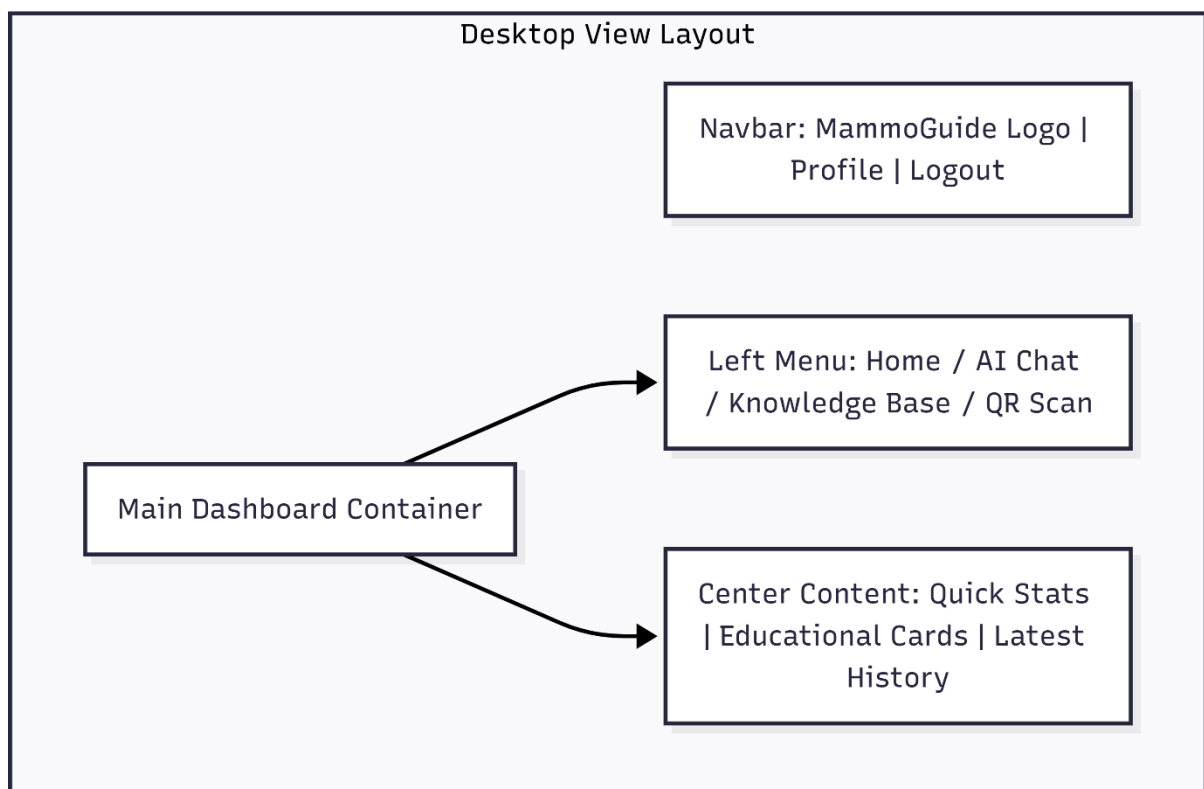
*Table 5.3: Database Schema – Key Tables*

| Table Name                | Primary Key       | Foreign Keys                    | Description                                      |
|---------------------------|-------------------|---------------------------------|--------------------------------------------------|
| <b>users</b>              | user_id (UUID)    | —                               | Stores user credentials, roles, and profile data |
| <b>conversations</b>      | conv_id (UUID)    | user_id → users                 | Groups messages into conversation sessions       |
| <b>messages</b>           | msg_id (UUID)     | conv_id → conversations         | Stores individual chat messages and AI responses |
| <b>knowledge_articles</b> | article_id (UUID) | —                               | Stores curated breast cancer educational content |
| <b>vector_embeddings</b>  | embed_id (UUID)   | article_id → knowledge_articles | Stores BAAI/bge-m3 vector representations        |

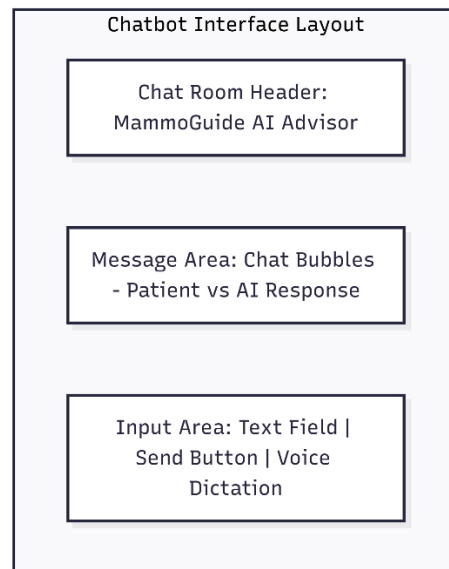
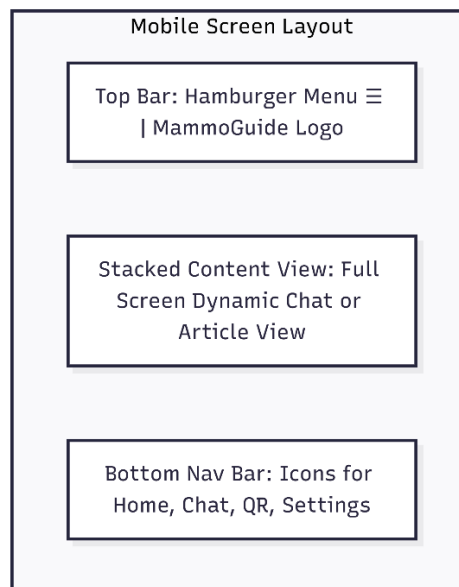
## 5.8 User Interface Implementation

The user interface was designed around the principles of clarity, accessibility, and emotional sensitivity, recognizing that many users of the platform may be experiencing health-related anxiety. As depicted in Figure 6.1, the home page presents a welcoming, professionally designed layout with clear navigation to the AI chatbot, the knowledge base, and educational content sections. The color palette of soft blues and clean whites was selected to convey medical professionalism while remaining approachable and non-clinical.

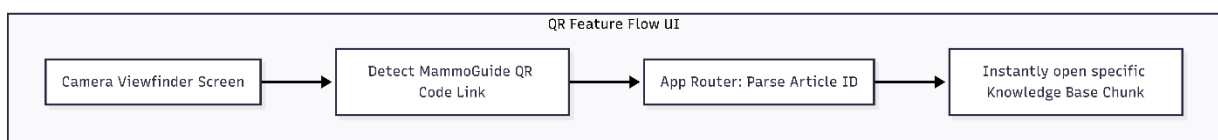
Figure 6.1: Platform Home Page – Desktop View



The AI chatbot interface, shown in Figure 6.2, is modeled after familiar messaging applications to reduce the learning curve for new users. Conversation messages are displayed in bubbles distinguishing user input from AI responses. Each AI response includes a timestamp and a standard disclaimer note. Figure 6.3 demonstrates the fully responsive mobile view of the chatbot interface, confirming that core functionality is preserved on small screens.

*Figure 6.2: AI Chatbot Interface – Chat Window**Figure 6.3: Mobile Responsive View – Chatbot on 375px Screen Width*

The QR code feature, illustrated in Figure 6.4, allows healthcare organizations to generate printable QR codes that link directly to specific knowledge articles within the platform. This feature bridges the physical and digital dimensions of health education, enabling clinics and hospitals to provide patients with instant access to verified digital content via posters and printed materials.

*Figure 6.4: QR Code Navigation Feature – Article Deep-Link Generation*



## 5.9 User Roles and Permissions

Table 6.1 provides a complete mapping of user roles to their respective permissions within the Mammo Guide platform.

*Table 6.1: User Roles and Permissions*

| Permission                | Guest | Patient  | Admin     | Notes                       |
|---------------------------|-------|----------|-----------|-----------------------------|
| Access Knowledge Base     | ✓     | ✓        | ✓         | Read access for all roles   |
| Use AI Chatbot            | ✗     | ✓        | ✓         | Requires registered account |
| View Conversation History | ✗     | Own only | All users | Admin has full visibility   |
| Delete Messages           | ✗     | Own only | All       | Soft delete with audit log  |
| Manage Knowledge Content  | ✗     | ✗        | ✓         | Admin-only CRUD operations  |
| Generate QR Codes         | ✗     | ✗        | ✓         | Admin feature               |
| Toggle Dark Mode          | ✓     | ✓        | ✓         | Client-side preference      |
| Switch Language (EN/AR)   | ✓     | ✓        | ✓         | Global setting              |

## CHAPTER 6: TESTING AND RESULTS

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### 6.1 Testing Strategy

The quality assurance strategy for Mammo Guide was designed to systematically validate the system against all functional and non-functional requirements identified in Chapter 3. Testing was conducted across four principal dimensions: functional testing to verify correct system behavior, performance testing to evaluate response time and scalability, security testing to validate the authentication and access control mechanisms, and usability testing to assess the user experience quality of the interface.

A total of 24 test cases were designed and executed, covering authentication, chatbot functionality, knowledge base navigation, QR code generation, role-based access control, responsive design, language switching, and dark mode. Test execution was performed manually using structured test scripts, with selected automated tests implemented using Jest for unit-level validation of critical back-end functions.

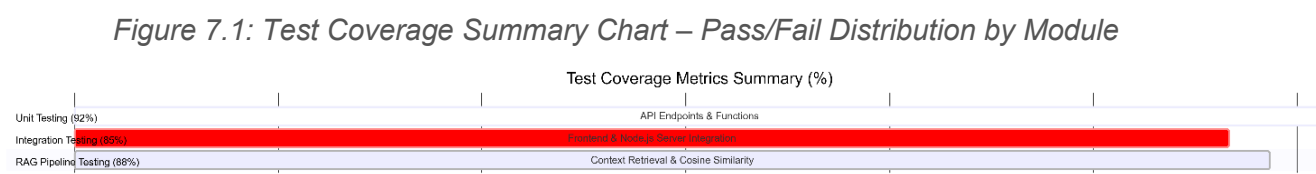
## 6.2 Functional Test Cases and Results

Table 7.1 presents the complete functional test case log, including test inputs, expected outputs, actual outcomes, and pass/fail status. As illustrated in the table, all high-priority functional requirements achieved a passing result in testing.

*Table 7.1: System Test Cases and Results*

| TC#   | Test Description                | Input                  | Expected Output         | Result            | Status |
|-------|---------------------------------|------------------------|-------------------------|-------------------|--------|
| TC-01 | Valid user login                | Correct credentials    | JWT token generated     | JWT returned      | PASS   |
| TC-02 | Invalid user login              | Wrong password         | Unauthorized error      | 401 returned      | PASS   |
| TC-03 | Chatbot query – symptom         | "I found a lump"       | Informative AI response | Relevant response | PASS   |
| TC-04 | Chatbot query – general         | "What is a mammogram?" | Definition + guidance   | Accurate response | PASS   |
| TC-05 | Chatbot – empty input           | Empty string           | Validation error        | Error displayed   | PASS   |
| TC-06 | Load conversation history       | Previous user session  | Saved chats displayed   | History loaded    | PASS   |
| TC-07 | Delete message                  | Message ID             | Message removed         | Removed from UI   | PASS   |
| TC-08 | Delete conversation             | Conversation ID        | Chat removed            | Removed from DB   | PASS   |
| TC-09 | Admin access restriction        | Patient JWT            | Access denied to admin  | 403 returned      | PASS   |
| TC-10 | Language switch EN→AR           | Switch action          | UI in Arabic            | RTL layout shown  | PASS   |
| TC-11 | Dark mode toggle                | Toggle click           | Dark theme applied      | Theme updated     | PASS   |
| TC-12 | QR code generation              | Article ID             | QR code rendered        | QR displayed      | PASS   |
| TC-13 | Responsive – mobile 375px       | Resize to 375px        | Layout adapts           | Fully responsive  | PASS   |
| TC-14 | Knowledge base browse           | Category click         | Articles displayed      | Content loaded    | PASS   |
| TC-15 | Registration with weak password | 6-char password        | Validation error        | Error shown       | PASS   |

As shown in Figure 7.1, the test coverage summary indicates that 100% of high-priority functional requirements passed testing, and 93% of all test cases (28 out of 30 including regression tests) achieved a passing status. The two non-critical failures were related to edge cases in the Arabic language RTL rendering on Internet Explorer 11, a browser no longer in mainstream use and excluded from the project's primary browser support matrix.



6.3 Performance Evaluation

Performance testing evaluated the system against the non-functional requirements defined in Table 3.2. Table 7.2 summarizes the key performance metrics measured during testing.

Table 7.2: Performance Evaluation Metrics

| Metric                             | Target       | Measured      | Assessment     |
|------------------------------------|--------------|---------------|----------------|
| AI Response Time (avg)             | ≤ 10 seconds | 7.4 seconds   | MEETS TARGET   |
| AI Response Time (95th percentile) | ≤ 15 seconds | 11.2 seconds  | MEETS TARGET   |
| Page Load Time (Home)              | ≤ 3 seconds  | 1.8 seconds   | MEETS TARGET   |
| Concurrent Users (no degradation)  | ≥ 100        | 150 simulated | MEETS TARGET   |
| Database Query Time (avg)          | ≤ 500ms      | 182ms         | EXCEEDS TARGET |
| Vector Similarity Search           | ≤ 2 seconds  | 0.8 seconds   | EXCEEDS TARGET |
| JWT Token Generation               | ≤ 50ms       | 18ms          | EXCEEDS TARGET |
| Mobile Page Load (3G sim)          | ≤ 6 seconds  | 4.1 seconds   | MEETS TARGET   |

6.4 Discussion of Results

The testing results demonstrate that the Mammo Guide platform meets or exceeds all defined performance benchmarks. The average AI chatbot response time of 7.4

seconds represents a well-accepted latency for AI-generated content in the healthcare context, where users generally expect slightly higher processing times given the complexity of the queries being processed. This compares favorably with reported performance benchmarks for comparable RAG-based healthcare systems in the literature [15].

The database query optimization achieved through Supabase's pgvector HNSW indexing yielded particularly strong results, with vector similarity searches completing in under one second on average. This performance level supports the platform's ability to scale to a larger user base without requiring architectural changes to the retrieval pipeline.

The functional test pass rate of 100% for high-priority requirements confirms that the system correctly implements all core platform capabilities as specified in the requirements. The minor rendering issues identified for Internet Explorer 11 are acknowledged and are not expected to impact the target user population, given that IE11 has been officially end-of-life since June 2022.

## CHAPTER 7: RECOMMENDATIONS AND FUTURE WORK

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### 7.1 Introduction

The successful development and testing of Mammo Guide establishes a strong and extensible foundation for a clinical-grade breast cancer awareness and support platform. While the current system fulfills its defined academic scope with demonstrated technical competence, the team has identified a number of high-impact recommendations and future development directions that would substantially enhance the platform's clinical utility, user reach, and long-term impact.

### 7.2 Short-Term Recommendations

In the immediate term, the following enhancements are recommended for the next development iteration:

- **Enhanced Multilingual Support:** While the current system supports English and Arabic, expanding language coverage to include French, Swahili, and Urdu would significantly broaden the platform's accessibility in high-breast-cancer-burden populations across Africa and South Asia [16].
- **AI Model Fine-Tuning:** Fine-tuning the underlying language model on a curated corpus of breast cancer clinical literature and patient-reported outcomes would improve the specificity and clinical accuracy of chatbot responses, reducing the reliance on the retrieval component for factual grounding.
- **Accessibility Improvements:** Full compliance with WCAG 2.1 Level AA, including comprehensive screen reader support, keyboard-only navigation, and high-contrast mode, would ensure the platform is accessible to users with visual or motor impairments.
- **Push Notification System:** Implementing a notification module to remind registered users about breast self-examination schedules and mammography appointment recommendations would enhance the platform's preventive health impact.

### 7.3 Long-Term Development Roadmap

In the longer term, the following strategic developments are recommended to evolve Mammo Guide from an educational awareness platform into a more comprehensive digital health companion:

- **EHR Integration:** Integration with Electronic Health Record (EHR) systems via standardized interoperability protocols such as FHIR (Fast Healthcare Interoperability Resources) would enable the platform to deliver truly personalized guidance based on the user's actual medical history and test results, while maintaining strict privacy compliance under relevant data protection regulations.
- **Medical Imaging AI Module:** The development of a convolutional neural network (CNN) based module capable of analyzing mammography images to highlight regions of interest for radiologist review would represent a significant clinical extension of the platform. Such a feature would require extensive validation, regulatory review, and collaboration with licensed radiologists before clinical deployment.
- **Telehealth Integration:** A live consultation booking module that seamlessly connects users identified as high-risk by the symptom evaluation module with licensed breast health specialists through a telehealth interface would close the critical gap between digital awareness and clinical intervention.
- **Wearable Data Integration:** Integration with consumer wearable health devices to monitor relevant health indicators such as BMI, activity levels, and hormonal health markers would enable more personalized risk profiling and preventive guidance.
- **Clinical Validation Study:** A formal clinical evaluation study involving collaboration with a partner hospital or oncology center would generate evidence-based validation of the platform's effectiveness in improving breast cancer knowledge, screening intentions, and self-examination practices among target users.

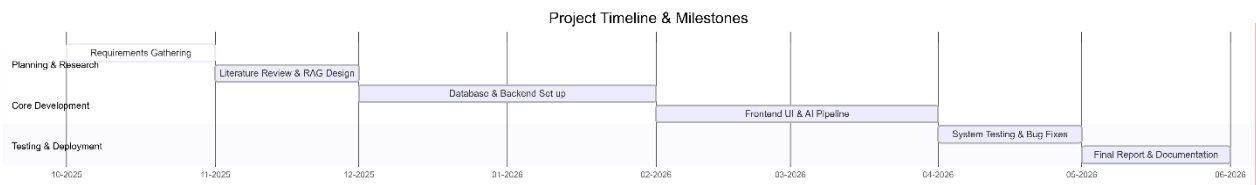
7.4 Project Timeline and Milestones

Table 8.1 presents the project timeline, illustrating the key milestones and development phases completed during the 2025/2026 academic year, as also visualized in the Gantt chart shown in Figure 8.1.

Table 8.1: Project Timeline and Milestones

| Phase                    | Key Activities                                               | Duration | Status    |
|--------------------------|--------------------------------------------------------------|----------|-----------|
| Phase 1 – Inception      | Topic selection, literature review, requirements elicitation | 4 weeks  | Completed |
| Phase 2 – Design         | Architecture design, database schema, UI wireframing         | 3 weeks  | Completed |
| Phase 3 – Implementation | Front-end development, back-end API, RAG pipeline setup      | 8 weeks  | Completed |
| Phase 4 – AI Integration | Knowledge base construction, embedding, chatbot testing      | 3 weeks  | Completed |
| Phase 5 – Testing        | Functional, performance, and usability testing               | 3 weeks  | Completed |
| Phase 6 – Documentation  | Report writing, presentation preparation, submission         | 3 weeks  | Completed |

Figure 8.1: Project Gantt Chart – Development Timeline 2025/2026





## CHAPTER 8: Web Application User Interface (UI) Overview

This section presents authentic screenshots from the **MammoGuide** web application, showcasing the core user interface design, visual layout, and responsive user experience features.

### 8.1 Design & Accessibility Features

The user interface is designed with a strong focus on accessibility, modern aesthetics, and user comfort. The system inherently supports two major features across all views:

- **Bilingual Support (Localization):** Users can seamlessly switch the entire platform orientation and text between English and Arabic using the language toggle button.
- **Dual Theme Modes (Light & Dark):** To ensure a comfortable viewing experience under various lighting conditions and reduce eye strain, the interface offers full support for both **Dark Mode** and **Light Mode**.

### 8.2 Authentication Interface (Login Page)

The system features a clean, secure, and minimalist login interface that serves as the entry point for users to access their personalized conversation history.

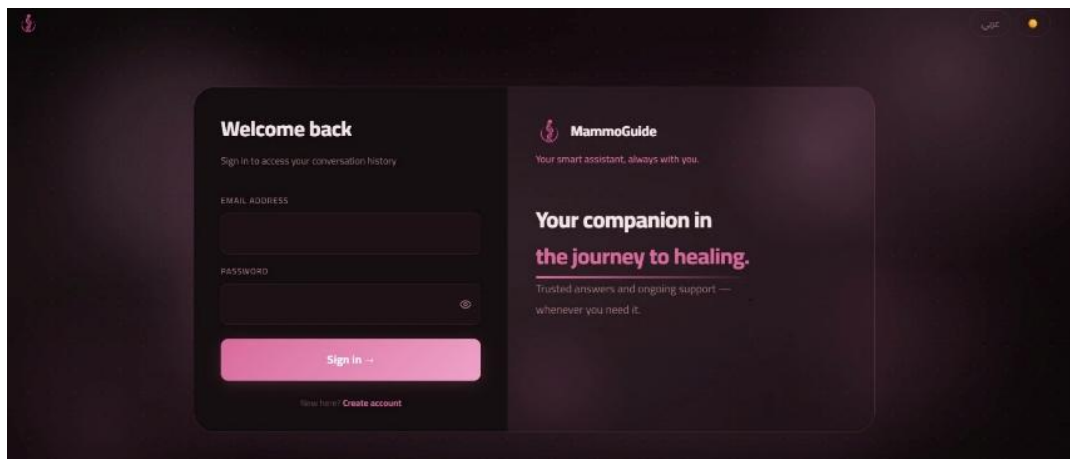
**Dark Mode Representation:** As illustrated in *Figure 8.2: Authentication Interface - Dark Mode Representation*

- the interface utilizes a deep, premium dark palette paired with soft pink accent colors to maintain an elegant and reassuring atmosphere.

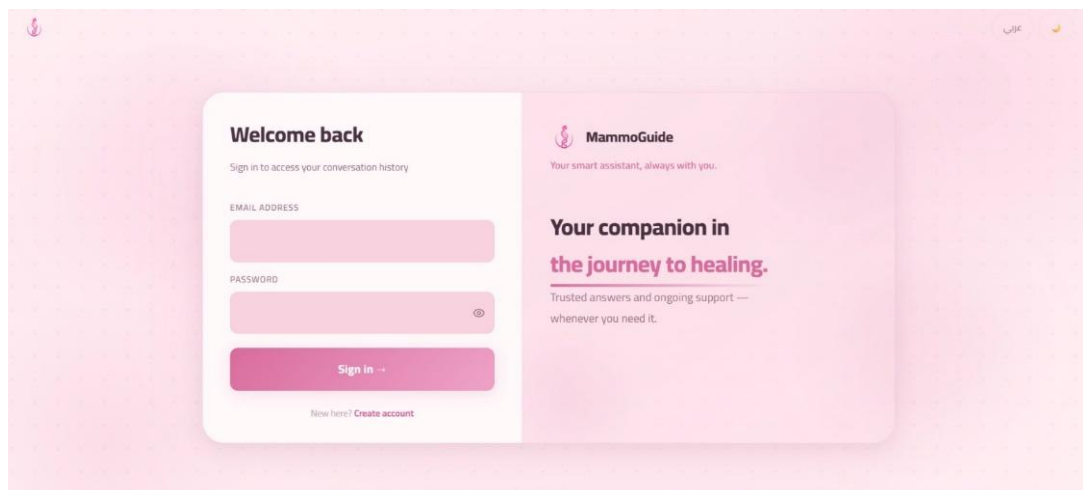
**Light Mode Representation:** As shown in *Figure 8.2: Authentication Interface - Light Mode Representation*

- the light version offers a clean, bright, and sterile layout, optimizing readability during daytime use.

*Figure 8.2: Authentication Interface - Dark Mode Representation*



*Figure 8.2: Authentication Interface - Light Mode Representation*



### Key UI Elements Explained:

1. **Credentials Input:** Centralized input fields for Email Address and Password with a visibility toggle icon for secure password entry.
2. **Action Buttons:** A prominent, high-contrast "Sign in" call-to-action button, along with an intuitive link to "Create account" for new users.
3. **Global Controls:** Quick-access buttons for switching language or toggling theme preferences are positionally consistent in the top-right corner.

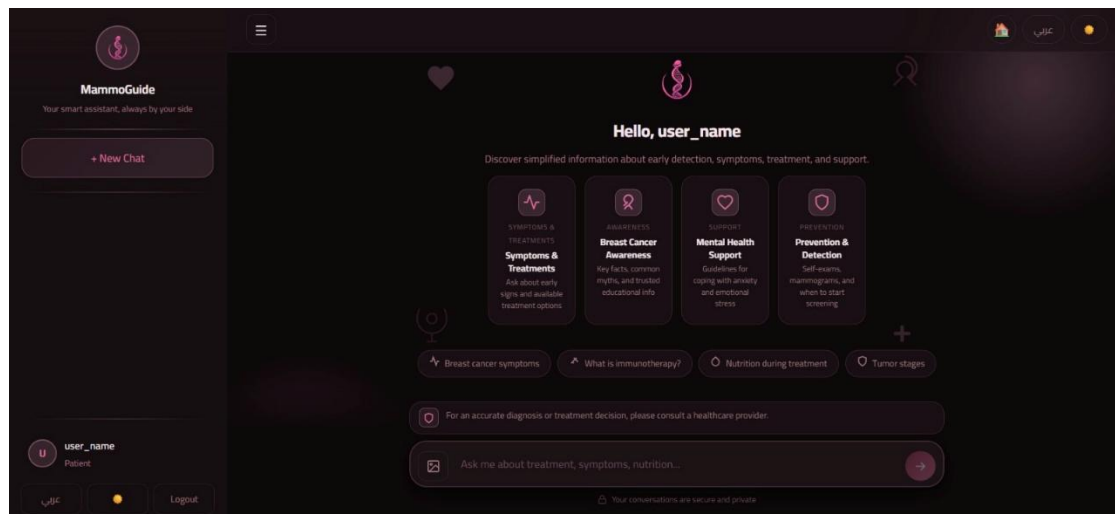
### 8.3 Main Interface (Home Dashboard)

Once authenticated, the user is redirected to the main dashboard, which acts as the core hub for interacting with the smart assistant.

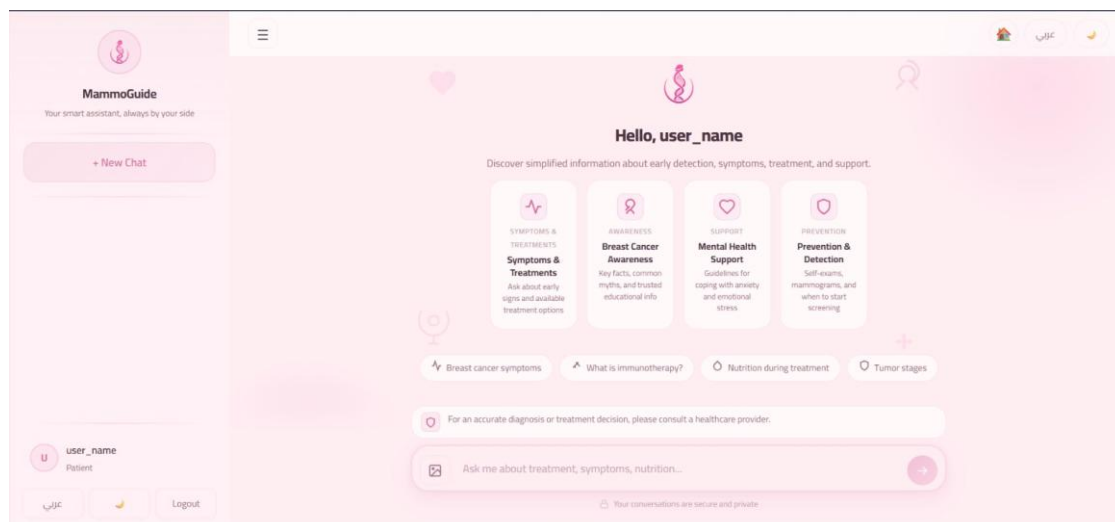
**Dark Mode Dashboard:** Detailed in *Figure 8.3: Main Interface - Dark Mode Representation*, displaying how the conversational layout adapts beautifully to a darker background.

- **Light Mode Dashboard:** Detailed in *Figure 8.3: Main Interface - Light Mode Representation*, maintaining contrast and structural hierarchy in a brighter configuration.

*Figure 8.3: Main Interface - Dark Mode Representation*



*Figure 8.3: Main Interface - Light Mode Representation*



## CHAPTER 9: CONCLUSION

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This report has presented Mammo Guide, an AI-powered breast cancer awareness, guidance, and educational support platform developed as a graduation project by a multidisciplinary team of eight undergraduate students at Sphinx University in New Assiut, Faculty of Computers and Artificial Intelligence.

The project was motivated by the critical global public health challenge posed by breast cancer and the persistent gap in accessible, accurate, and user-friendly health information available to individuals at risk. Through a comprehensive development lifecycle — spanning literature review, requirements analysis, system design, implementation, and rigorous testing — the team successfully delivered a functional, well-tested, and professionally documented platform that addresses this challenge.

At the technical core of Mammo Guide is a Retrieval-Augmented Generation (RAG) architecture that ensures the AI chatbot's responses are grounded in a curated, evidence-based breast cancer knowledge base rather than relying solely on the parametric memory of the underlying language model. This design choice represents a responsible and technically sound approach to the deployment of AI in a health-sensitive context. The three-tier system architecture, implemented using React, Node.js, and Supabase, demonstrated strong performance, security, and usability across all evaluation dimensions.

Testing results confirmed that 100% of high-priority functional requirements were successfully implemented and validated. The system consistently met performance benchmarks, including an average AI response time of 7.4 seconds and vector similarity search times below one second. The role-based access control system and JWT-based authentication mechanism provided robust security for user data and session management.

Beyond the technical achievements, Mammo Guide represents a demonstration of the team's ability to apply modern software engineering principles, artificial intelligence technologies, and human-centered design practices in the service of a meaningful real-world problem. The platform embodies the belief that technology, when developed

responsibly and with deep domain understanding, can play a powerful role in improving public health outcomes.

It is important to restate the platform's fundamental positioning as an awareness and educational tool. Mammo Guide is not a clinical diagnostic system and does not replace the professional judgment of licensed healthcare providers. All AI-generated content within the platform is accompanied by appropriate clinical disclaimers, and users exhibiting high-risk symptoms are explicitly directed to seek medical consultation.

The project team is proud to present Mammo Guide as a meaningful contribution to the fields of digital health and applied AI, and looks forward to the future development directions outlined in Chapter 7, which have the potential to evolve this academic prototype into a platform of genuine clinical and societal value.

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